Inflation Dynamics

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Abstract

This is my abstract!

1. Introduction

I am a book about … something! Wikipedia has information about books: hover over the link for more information.

> **Note**
>
> Books are usually written on paper … But Jupyter Book can create *websites*!

If you sold 100 books at $10 per book, you'd have $1000 dollars according to (1). If instead you publish your Jupyter Book to the web for free, you'd have $0 dollars!

$$x \times y = z \tag{1}$$

Sometimes when reading it is helpful to foster a *tranquil* environment. The image in Figure 1 would be a perfect spot!

Figure 1: A photograph of some beautiful mountains to look at whilst reading.

Computational Notebooks

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2. Inflation Dynamics

2.a. I. Mean Processes

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = fc_t + \mu_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 fc_t + \mu_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 fc_t + \mu_{t+1}$
- **ARX(2,2)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 fc_t + \phi_2 fc_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2.a.i. *I.a. OLS Results:*

I report OLS estimation results of three mean models[1]:

ARX(1,1):

$$\begin{aligned} \hat{\pi}_{t+1} &= 0.0720 + 0.2881\,\pi_t + 0.7508\,fc_t \\ &= (0.087) \quad (0.114) \qquad (0.168) \end{aligned} \tag{2}$$

ARX(2,1):

$$\begin{aligned} \hat{\pi}_{t+1} &= 0.0792 + 0.2793\,\pi_t + 0.0728\,\pi_{t-1} + 0.6661\,fc_t \\ &= (0.087) \qquad (0.102) \qquad (0.127) \qquad\quad (0.257) \end{aligned} \tag{3}$$

ARX(2,2):

$$\begin{aligned} \hat{\pi}_{t+1} &= 0.0761 + 0.2880\,\pi_t + 0.0799\,\pi_{t-1} + 0.4720\,fc_t + 0.1793\,fc_{t-1} \\ &= (0.088) \qquad (0.109) \qquad (0.127) \qquad\quad (0.440) \qquad\quad (0.295) \end{aligned} \tag{4}$$

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma,\ \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	$-$	$-$	$-$	$-$	$-$
ARX(1,1)	$(\min \pi_t,\ \max \pi_t)$	$(-0.999,\ 0.999)$	$-$	$(-10,\ 10)$	$-$
ARX(2,1)	$(\min \pi_t,\ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	$-$
ARX(2,2)	$(\min \pi_t,\ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	$(-10,\ 10)$

2.b. II. GARCH Model Summarization

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC

[1]Standard errors are heteroskedasticity and autocorrelation robust (HAC) using 12 lags and without small sample correction.

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew t	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew t	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

2.c. III. BEGE Density

BEGE density function $f_{BEGE}(\mu \texttt{ mid } p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

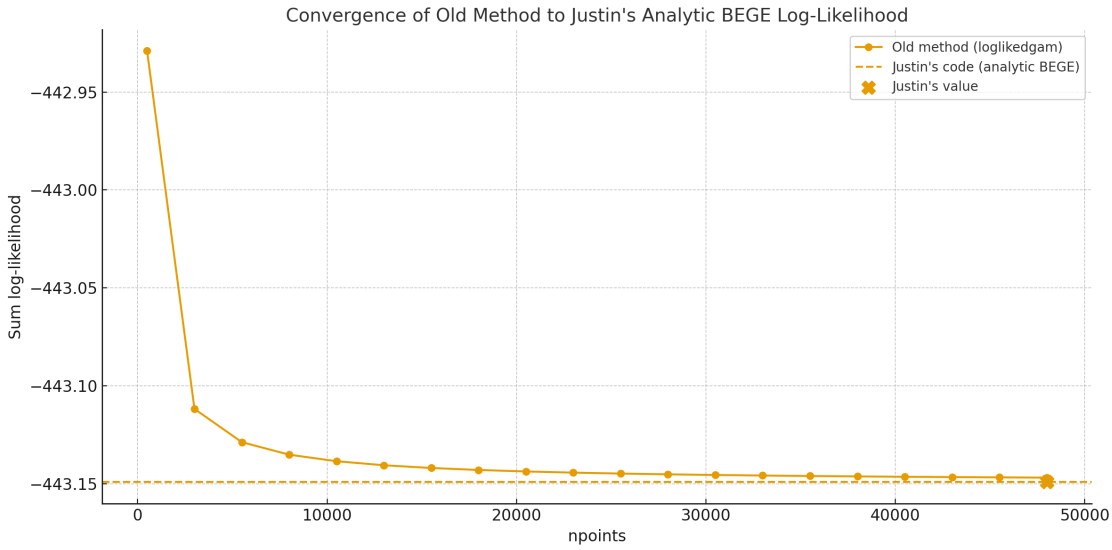
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \qquad n = 7, \qquad \sigma_p = 0.8, \qquad \sigma_n = 1.2. \tag{5}$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function.*

The old numerical method was evaluated over a wide range of grid resolutions (up to $50,000$ points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

2.d. *IV. BEGE GARCH Family*

2.d.i. *IV.a. Constant BEGE:*

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n \end{aligned} \tag{6}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1). \tag{7}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

2.d.ii. *IV.b. Symmetric BEGE:*

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \, \omega_p - \sigma_n \, \omega_n, \end{aligned} \tag{8}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{9}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned} p_t &= p_0 + \rho\, p_{t-1} + \frac{\phi^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho\, n_{t-1} + \frac{\phi^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{10}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

2.d.iii. *IV.c. Bad and Good Symmetric GARCH Model:*

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p}{2\sigma_p^2}\left(u_{t-1}\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n}{2\sigma_n^2}\left(u_{t-1}\right)^2 \end{aligned} \tag{11}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

2.d.iv. *IV.d. Inflation and Deflation Model:*

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{12}$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

2.d.v. *IV.e. Full BEGE:*

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \, \omega_p - \sigma_n \, \omega_n, \end{aligned} \tag{13}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{14}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p \, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n \, n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2 \end{aligned} \tag{15}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,
- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$ where $\mathrm{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$.

2.e. V. Result Summary

Overall, most of the estimations with random initial values converge. I randomly sampled $100,000$ starting values for the constant BEGE model, and sampled $10,000$ for symmetric BEGE because it is more time consuming. I report the best estimation below.